Generate, Retrieve, or Calculate?

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

Recall and choose your starting point

> Record your starting explanation for Generate, Retrieve, or Calculate? before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Privacy by Choosing Less

Without opening the old lesson, explain one key idea from Privacy by Choosing Less. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Missing and Unbalanced Examples

Without opening the old lesson, explain one key idea from Missing and Unbalanced Examples. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Similarity

Without opening the old lesson, explain one key idea from Similarity. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

Complete the supported Generate, Retrieve, or Calculate? trace and record the first step you can explain without help.

Use W2: Separate a correct sum from a correct answer for **Generation retrieval** and **calculation**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Generate, Retrieve, or Calculate? explanation against the supplied facts before discussing it.

Use the worked case for **Generation retrieval** and **calculation**. Proposed explanation: “Retrieving a record proves it is current”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Trace a new library workshop

Current fictional record B lists a library workshop on Friday in Room 5: five tables with six places each, with four places reserved for helpers. B replaces old record A, which lists Room 3.

A draft says “Thursday in Room 3, with 30 places remaining and a free bus home.” **Retrieve** the current day and room, **calculate** the non-helper places, write a corrected factual sentence and identify the detail that cannot be supported by B.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Generate, Retrieve, or Calculate? target to a trace and a justified conclusion.

Three tasks are: invent a school motto, find the exact fire-drill meeting point in a supplied handbook, and divide 84 stickers among 7 teams. Choose **generation**, **retrieval** or **calculation** for each. Which **output**s need checking?

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Generate, Retrieve, or Calculate? case and show the deciding evidence.

A learner needs a new poem, 18+7, and a passage from the class handbook.

Choose the operation for each.

Show your **trace**, **calculation** or **evidence**

Write or draw your reasoning and evidence.

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

Choose a method for each task: invent a dragon story; find the library's posted hours; compute 7 plus 5. Explain why the methods differ.

My independent answer and reason

Write or draw your reasoning and evidence.

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Generate, Retrieve, or Calculate? adds to Helpful Classroom Sorter and keep the evidence.

Label every sorter component as **generate**, **retrieve**, **calculate**, fixed rule, learned **prediction**, or **human decision**.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Three tasks are: invent a school motto, find the exact fire-drill meeting point in a supplied handbook, and divide 84 stickers among 7 teams. Choose **generation**, **retrieval** or **calculation** for each. Which **output**s need checking?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Name the operation and its evidence

Name the operation and its evidence. Apply the supplied Generate, Retrieve, or Calculate? facts and keep your reasoning.

For each task name the main operation and a suitable check: find the room listed in N2; work out six tables of four places minus three reserved places; suggest an inviting title for the event.

Can an application combine these operations?

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Separate a correct sum from a correct answer

> Separate a correct sum from a correct answer. Apply the supplied Generate, Retrieve, or Calculate? facts and keep your reasoning.

A calculator correctly returns 24 for 6 × 4. A pupil reports this as the number of non-helper places available. Explain the problem and write the correct expression, result and meaning.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Inspect a record and a draft

> Inspect a record and a draft. Apply the supplied Generate, Retrieve, or Calculate? facts and keep your reasoning.

N1 lists Room 2. N2 explicitly replaces N1 for this event and lists Wednesday in Room 4. A draft says Thursday in Room 2 with free snacks. Identify the supported corrections and the unsupported detail.

Explain why finding N1 successfully would not settle the current-room question.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Trace a new library workshop

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Trace a new library workshop. Apply the supplied Generate, Retrieve, or Calculate? facts and keep your reasoning.

Current fictional record B lists a library workshop on Friday in Room 5: five tables with six places each, with four places reserved for helpers. B replaces old record A, which lists Room 3.

A draft says “Thursday in Room 3, with 30 places remaining and a free bus home.” **Retrieve** the current day and room, **calculate** the non-helper places, write a corrected factual sentence and identify the detail that cannot be supported by B.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Do not infer a method from its output

> Do not infer a method from its output. Apply the supplied Generate, Retrieve, or Calculate? facts and keep your reasoning.

Two applications both display “26 places remain.” Only one provides record B and a **trace** showing 5 × 6 – 4. What can and cannot be concluded from their matching sentences?

Describe how you would check the numerical claim without relying on either application’s assurance.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	

Evidence	My record
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Generate, Retrieve, or Calculate?

> A fictional craft invitation combines three different operations. Explain the labelled step before checking the changed condition.

A fictional craft invitation combines three different operations.

Approved record C: Tuesday, Hall 4, three tables with four learner places each.

Two places are reserved for helpers; no snack information is supplied.

AI ACADEMY / AI-1/25

Generate, Retrieve, or Calculate?

Start with the supplied facts
Follow each arrow. Ask which fact allows the next step.

1. Retrieve facts

Read Tuesday, Hall 4 and 3 x 4 places from approved C.

2. Calculate

12 total places minus 2 reserved =10 learner places.

3. Generate wording

Draft a friendly invitation using only those supported details.

4. Check claims

A sentence promising snacks has no supporting record.

CHANGE ONE THING
Reserve four places instead of two, keeping the room and day unchanged.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Label each arrow with its operation. Can the system add snacks because invitations often mention them?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Use the sorter’s approved catalogue for facts, explicit **calculation**s for scores and checked prose for explanations.

Project portfolio milestone

> Keep this week’s input, deciding step, result and limit together in the Helpful Classroom Sorter record.

Project: Helpful Classroom Sorter

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the sorter’s approved catalogue for facts, explicit **calculation**s for scores and checked prose for explanations.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional craft invitation combines three different operations.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? A sentence promising snacks has no supporting record.

> 1. **Retrieve** facts

2. **Calculate**

3. **Generate** wording

4. Check claims

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Check claims A sentence promising snacks has no supporting record.

Retrieval copies relevant source information; **calculation** follows arithmetic; **generation** proposes wording. These jobs can cooperate, but one does not validate the others.

A friendly **generate**d sentence can still contain an invented detail. Keep the approved source beside the draft and check each factual claim separately.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Reserve four places instead of two, keeping the room and day unchanged. Which conclusion is supported by the supplied facts?

1. Eight learner places remain.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial **trace**. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Eight learner places remain.

> 3×4−4=8; this changes capacity for learners, not the **retrieve**d date or location.

Repair a missing building block

Review Privacy by Choosing Less (ai-1/24). Goal: Justify which information a stated task needs and remove unnecessary fields.

Short practice: For the original task, does adding an anonymous-looking pupil code improve the repair count?. Return to this week and explain what you changed.

Review Missing and Unbalanced Examples (ai-1/22). Goal: Name the target collection, selection list and cases actually assessed. Short practice: Would testing only dim plain cards close every current gap?

Point to the target groups.. Return to this week and explain what you changed.

Review Similarity (ai-1/19). Goal: **Calculate** similarity using explicitly selected features and a stated scoring rule. Short practice: Point to the matches supporting C's new score and explain why there is no tie..

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

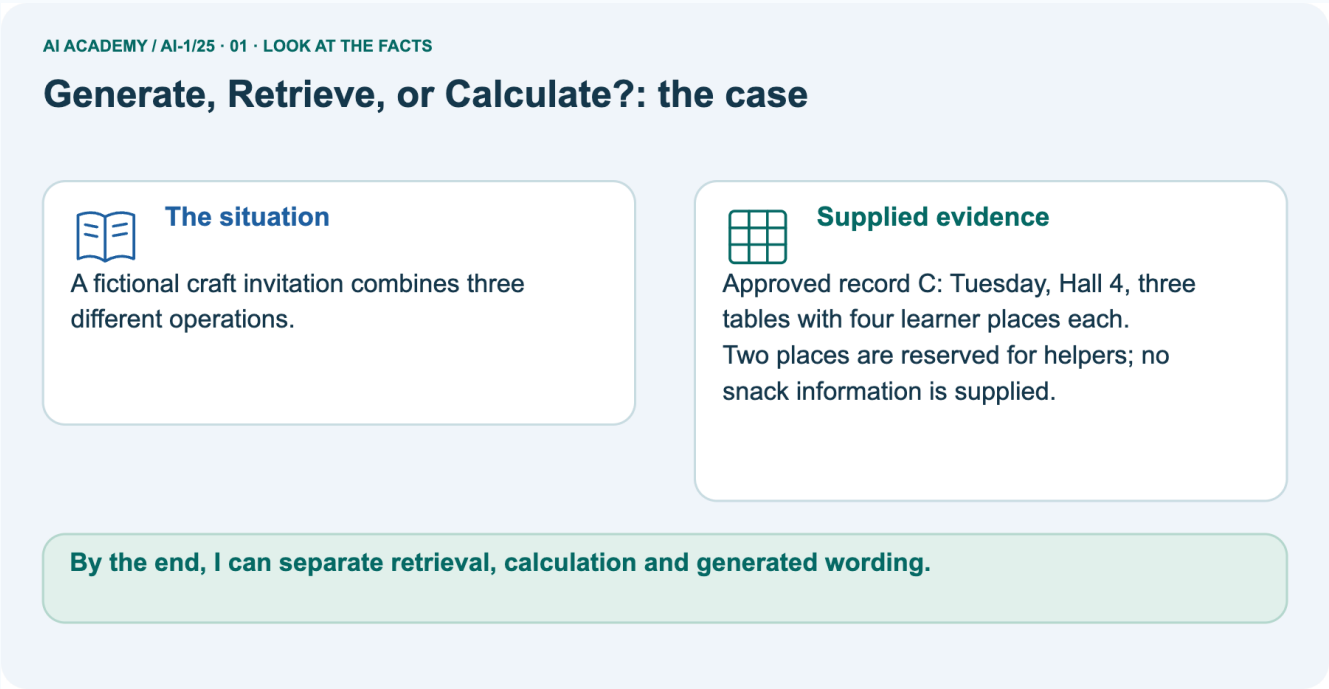
Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

Visual concept journey

> Explain each Generate, Retrieve, or Calculate? picture from its labels, then record a relevant limit.

01. Generate, Retrieve, or Calculate?: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

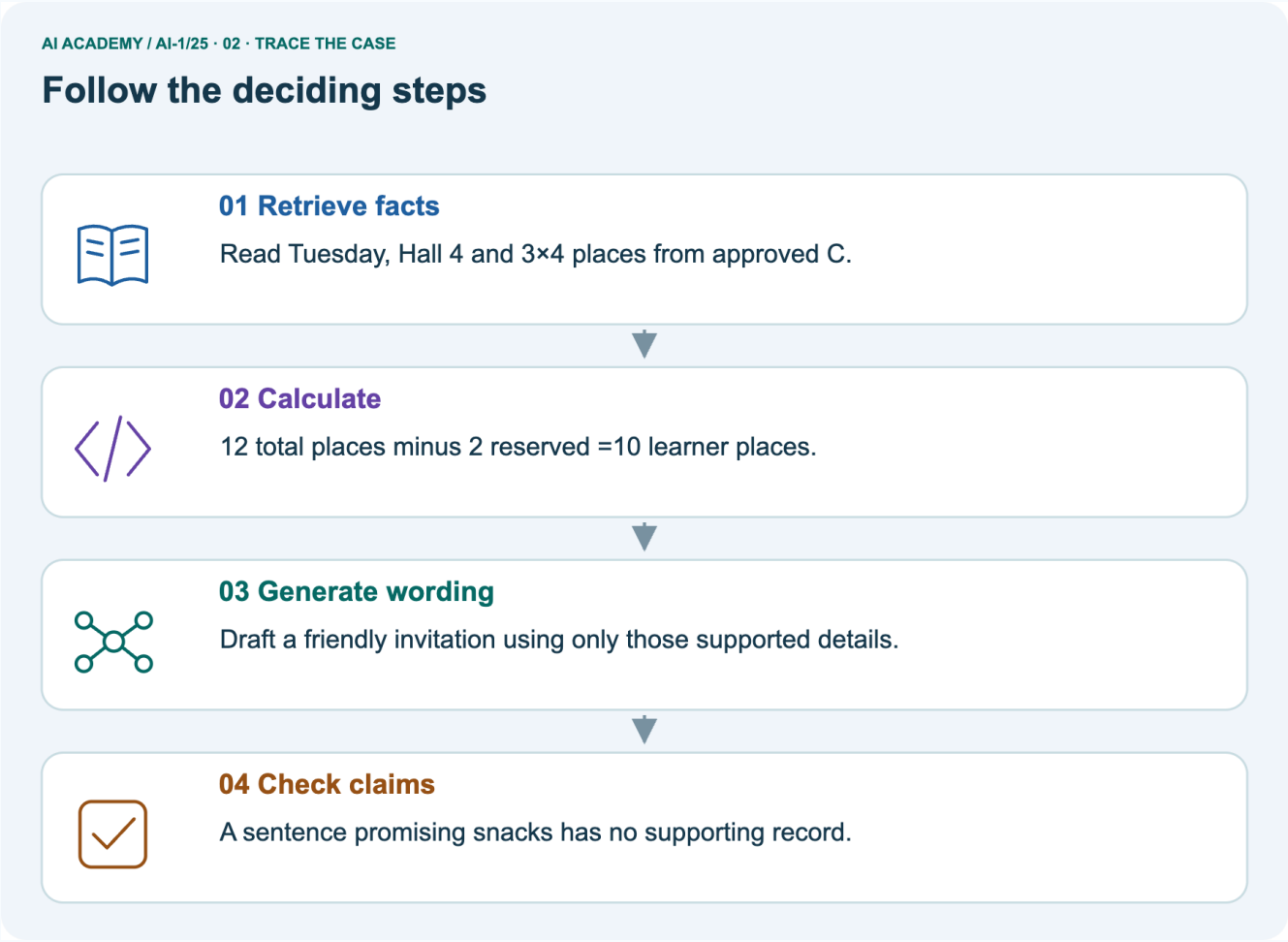
A fictional craft invitation combines three different operations. Supplied facts: Approved record C: Tuesday, Hall 4, three tables with four learner places each.

Two places are reserved for helpers; no snack information is supplied.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



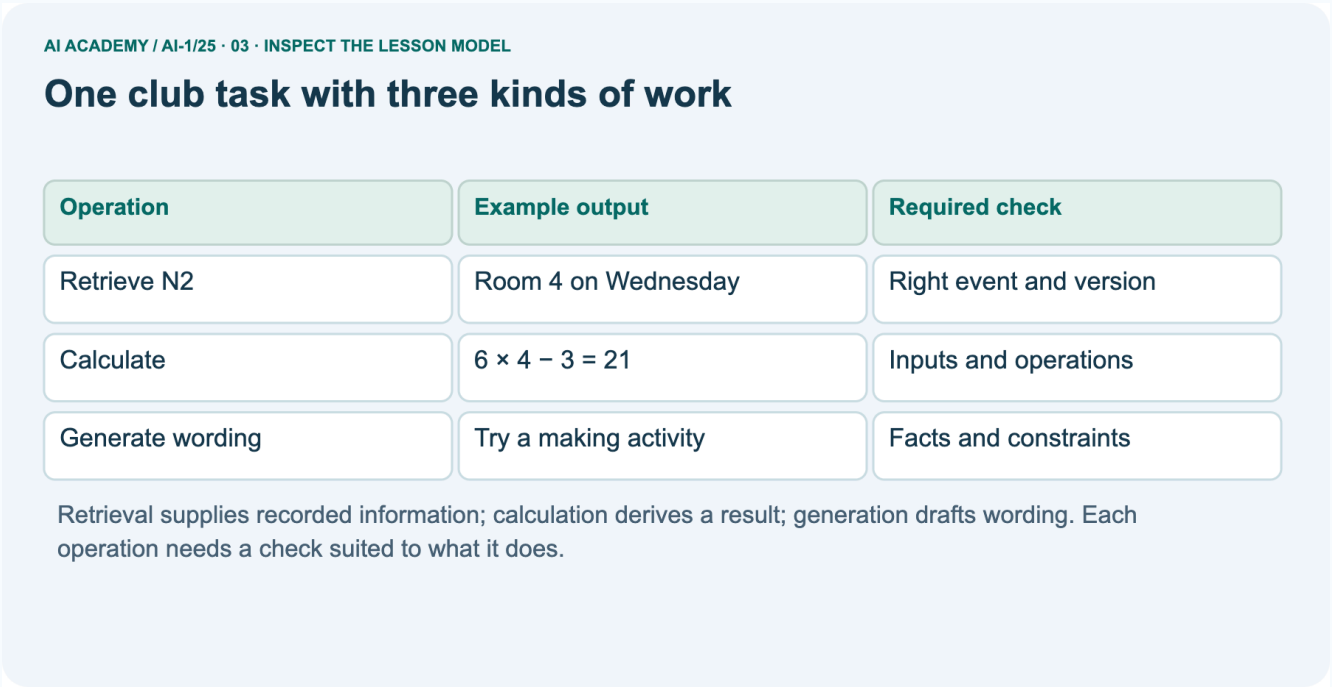
Follow the labelled picture with the matching explanation. Open the vector diagram

Retrieve facts: Read Tuesday, Hall 4 and 3×4 places from approved C. → **Calculate**: 12 total places minus 2 reserved =10 learner places. → **Generate** wording: Draft a friendly invitation using only those supported details. → Check claims: A sentence promising snacks has no supporting record.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. One club task with three kinds of work



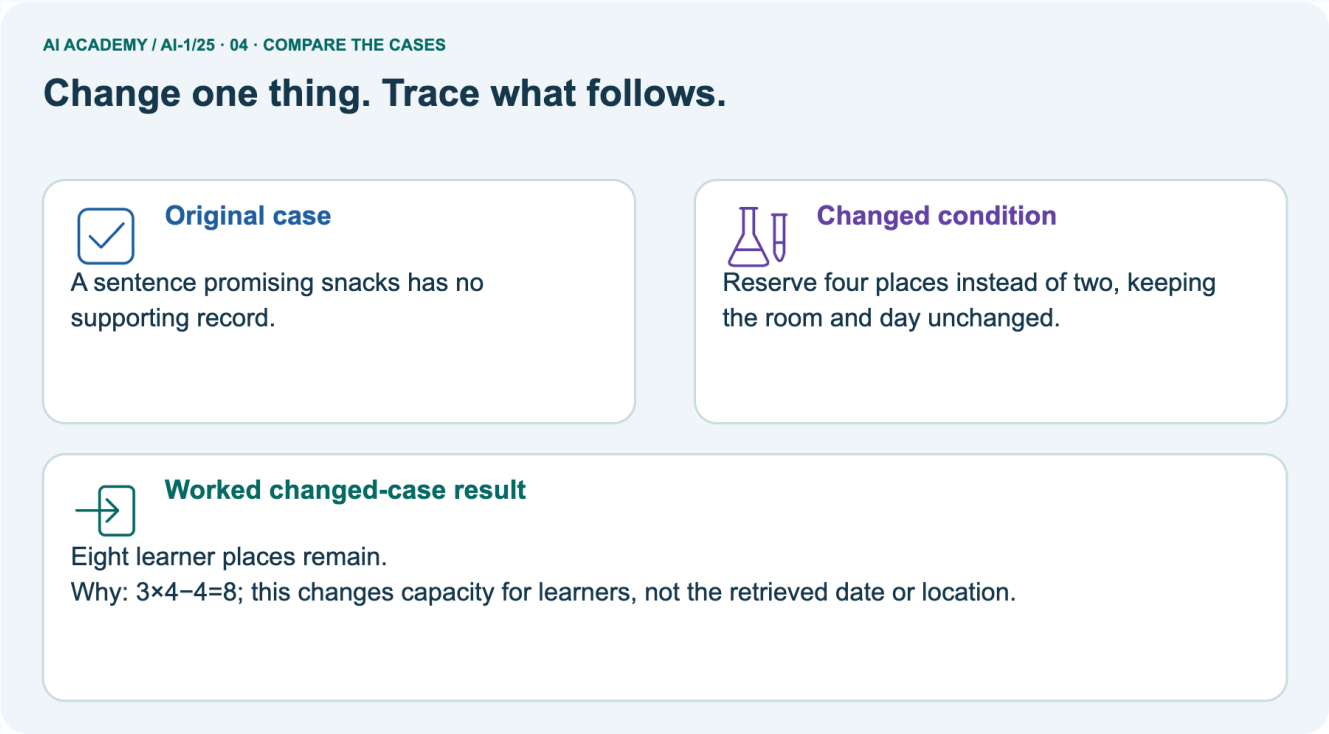
Follow the labelled picture with the matching explanation. Open the vector diagram

One club task with three kinds of work. **Retrieve** N2 | Room 4 on Wednesday | Right event and version. **Calculate** | 6 × 4 − 3 = 21 | **Input**s and operations. **Generate** wording | Try a making activity | Facts and constraints

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

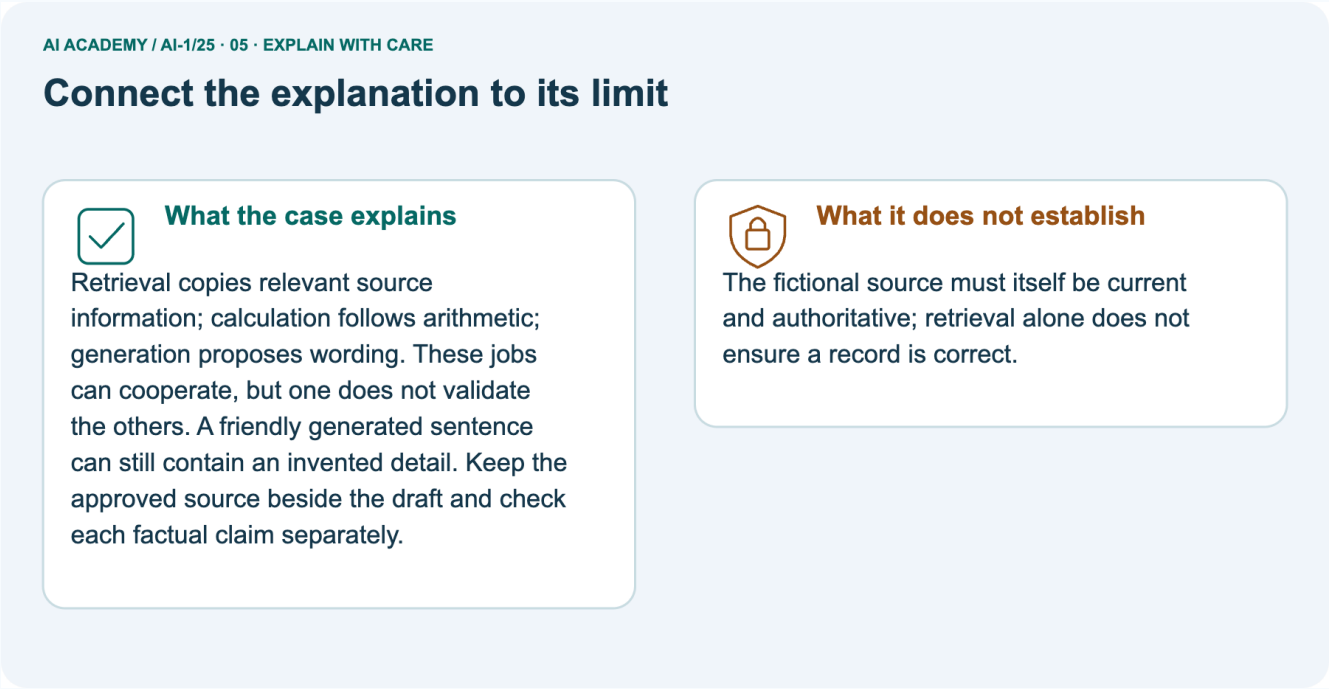
Original case: A sentence promising snacks has no supporting record. Changed condition: Reserve four places instead of two, keeping the room and day unchanged. Expected result: Eight learner places remain.

Reason: 3×4−4=8; this changes capacity for learners, not the **retrieve**d date or location.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Retrieval copies relevant source information; **calculation** follows arithmetic; **generation** proposes wording. These jobs can cooperate, but one does not validate the others.

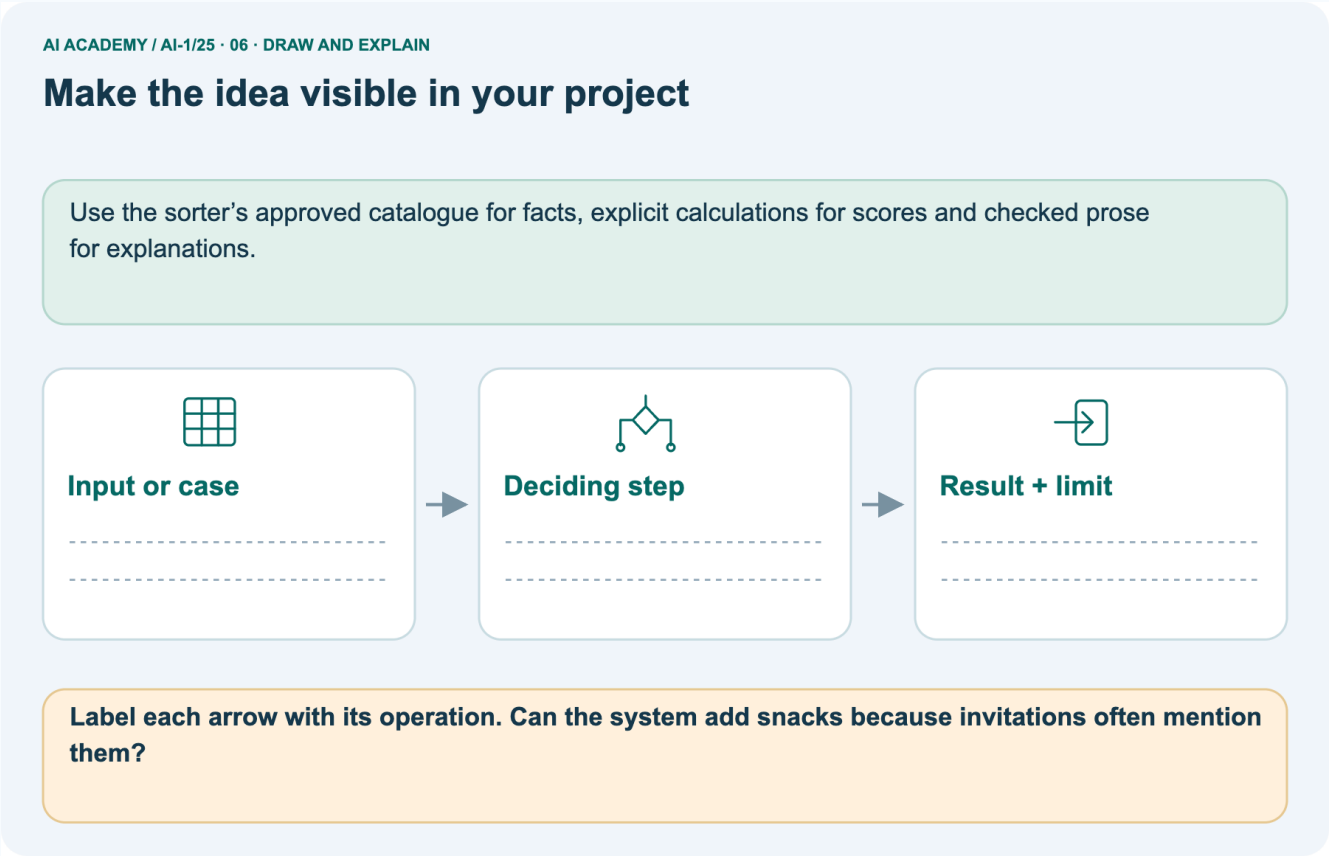
A friendly **generate**d sentence can still contain an invented detail. Keep the approved source beside the draft and check each factual claim separately.

Limit: The fictional source must itself be current and authoritative; **retrieval** alone does not ensure a record is correct.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Use the sorter's approved catalogue for facts, explicit **calculation**s for scores and checked prose for explanations. Draw your **input**, deciding step, result and limit. Supported question: Label each arrow with its operation.

Can the system add snacks because invitations often mention them?

Draw your project **trace** and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.